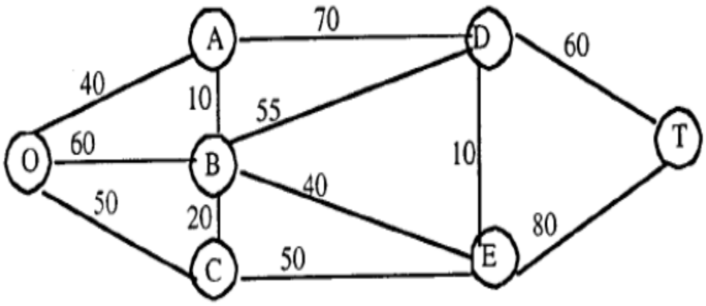
題號 9.3-2 (a)(b)

10.3-2.

(a)



(b)

n	Solved Nodes Directly Connected to Unsolved Nodes	Closest Connected Unsolved Node	Total Distance Involved	nth Nearest Node	Minimum Distance	Last Connection
1	O	A	40	A	40	OA
2, 3	O A	C B	50 40+10 = 50	C B	50 50	OC AB
4	A B C	D E E	40+70 = 110 50+40 = 90 50+50 = 100	E	90	BE
5	A B E	D D D	40+70 = 110 50+55 = 115 90+10 = 100	D	100	ED
6	D E	T T	100+60 = 160 90+80 = 170	T	160	DT

The shortest path from the origin to the destination is O → A → B → E → D → T, with a total distance of 160 miles.

題號 9.3-6 (b)

10.3-6.

(a) The flying times play the role of "distances."

(b) Shortest path: SE → C → E → LN, with total flight time 11.3

n	Solved Nodes Directly Connected to Unsolved Nodes	Closest Connected Unsolved Node	Total Distance Involved	*n*th Nearest Node	Minimum Distance	Last Connection
1	SE	C	4.2	C	4.2	SE-C
2	SE C	A F	4.6 4.2+3.4 = 7.6	A	4.6	SE-A
3	SE C A	B F E	4.7 4.2+3.4 = 7.6 4.6+3.4 = 8	B	4.7	SE-B
4	A B C	E E F	4.6+3.4 = 8 4.7+3.2 = 7.9 4.2+3.4 = 7.6	F	7.6	C-F
5	A B C F	E E E LN	4.6+3.4 = 8 4.7+3.2 = 7.9 4.2+3.5 = 7.7 7.6+3.8 = 11.4	E	7.7	C-E
6	A B F E	D D LN LN	4.6+3.5 = 8.1 4.7+3.6 = 8.3 7.6+3.8 = 11.4 7.7+3.6 = 11.3	D	8.1	A-D
7	D E F	LN LN LN	8.1+3.4 = 11.5 7.7+3.6 = 11.3 7.6+3.8 = 11.4	LN	11.3	E-LN

7. 找出圖 8.22 之網路的最小擴充樹。

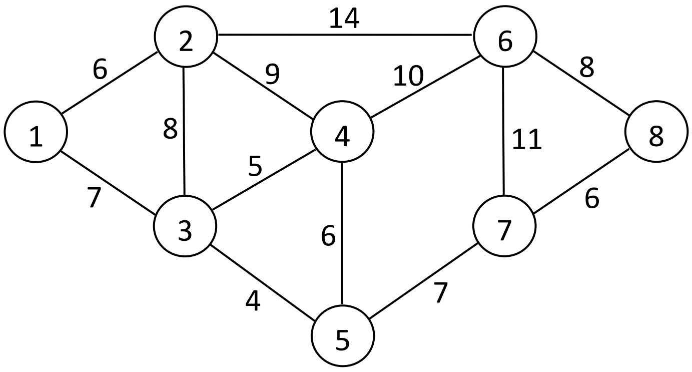


圖 8.22

Sol. 應用最小擴充樹的演算法，可得以下的求解程序（參見圖 f）:

1. 選擇長度最短的弧 (3,5) 。
2. 因節點 4 與目前的連接圖最近，所以加入弧 (3,4) 。
3. 繼續求解程序，會依序加入 (1,3) 、 (1,2) 、 (5,7) 、 (7,8) 、 (6,8) 。

所得到最小擴充樹的總長度為 43。

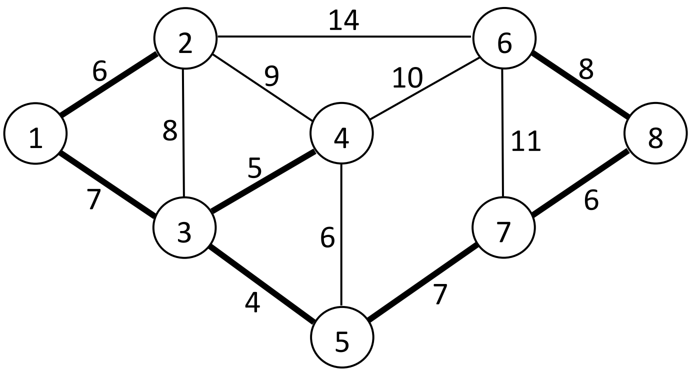


圖 f

14. 考慮圖 8.26 的網路。決定由節點 1 至節點 9 的最大流量以及各弧的最佳流量。

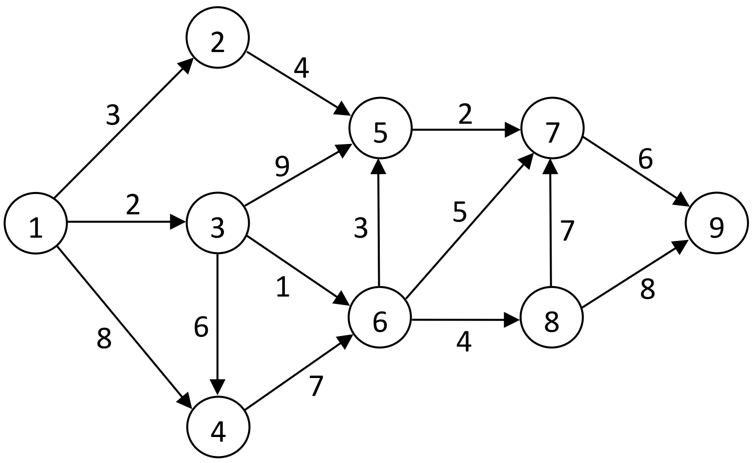


圖 8.26

Sol. 應用最大流量問題演算法可得圖 n 的求解過程。表 G 列出以上所找出的所有路徑。欲決定各弧的最佳流量，我們可比較圖 n 與圖 8.26，兩圖之各弧的流量差異即為各弧的最佳流量，結果如圖 o 所示。

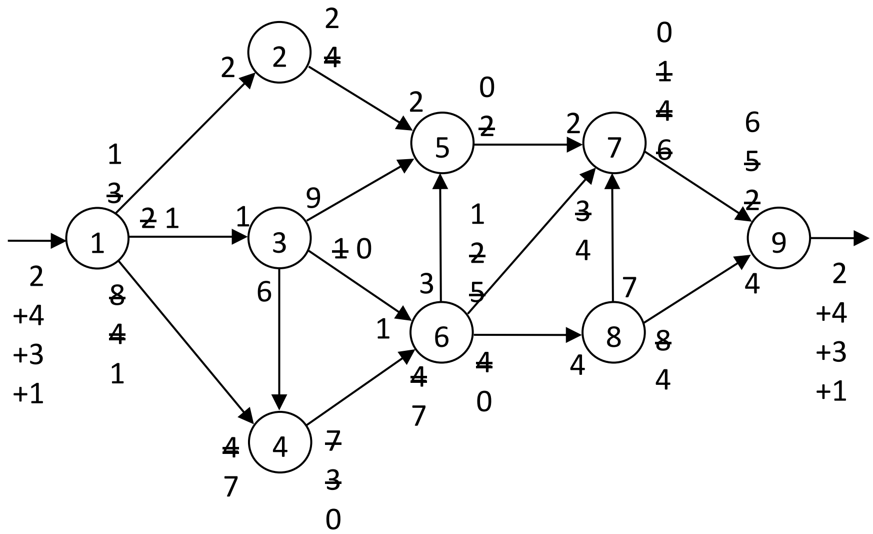


圖 n

表 G

路徑	流量
1→2→5→7→9	2
1→4→6→8→9	4
1→4→6→7→9	3
1→3→6→7→9	1
合計	10

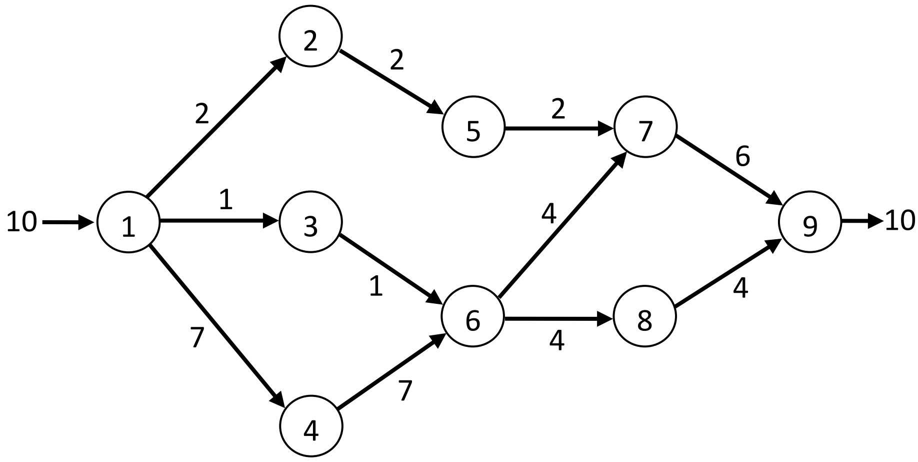
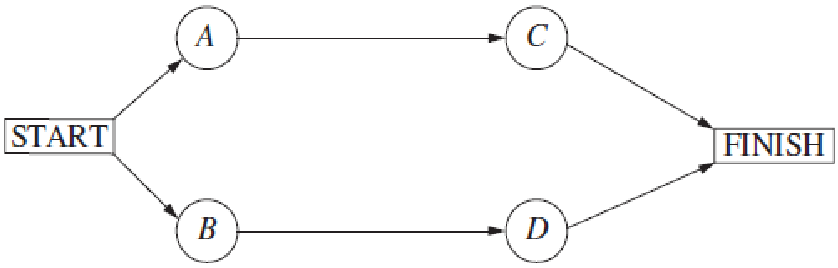


圖 o

<u>題號 9.8-1</u>

注意：課本圖有錯，正確如下：



10.8-1.

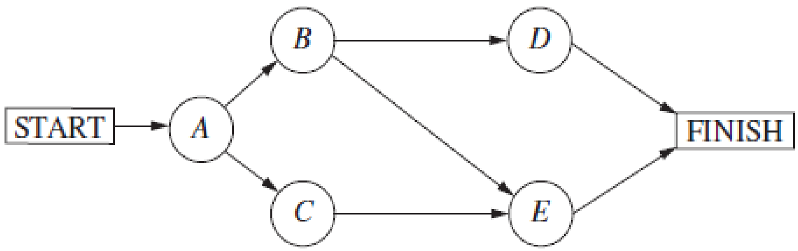
		Length of Path	
Activity to Crash	Crash Cost	$A-C$	$B-D$
		14	16
B	\$5,000	14	15
B	\$5,000	14	15
D	\$6,000	14	14
C	\$4,000	13	14
D	\$6,000	13	13
C	\$4,000	12	13
D	\$6,000	12	12

題號 9.8-2 (d)

(d) Let x_j be the reduction in the duration of activity j due to crashing for $j = A, B, C, D$. Also let y_j denote the start time of activity j for $j = C, D$ and y_{FINISH} the project duration.

$$
\begin{array}{ll}
\text{minimize} & C = 5000x_A + 5000x_B + 4000x_C + 6000x_D \\
\text{subject to} & x_A \le 3, x_B \le 2, x_C \le 2, x_D \le 3 \\
 & y_C \ge 0 + 8 - x_A \\
 & y_D \ge 0 + 9 - x_B \\
 & y_{\text{FINISH}} \ge y_C + 6 - x_C \\
 & y_{\text{FINISH}} \ge y_D + 7 - x_D \\
 & y_{\text{FINISH}} \le 12 \\
\text{and} & x_A, x_B, x_C, x_D, y_C, y_D, y_{\text{FINISH}} \ge 0
\end{array}
$$

題號 9.8-3 (a)　　**注意：課本圖有錯，正確如下：**



10.8-3.

(a) $7,834 is saved by the new plan given below.

Activity to Crash	Crash Cost	Length of Path		
		$A-B-D$	$A-B-E$	$A-C-E$
		10	11	12
C	$1,333	10	11	11
E	$2,500	10	10	10
D & E	$4,000	9	9	9
B & C	$4,333	8	8	8

Activity	Duration	Cost
A	3 weeks	$54,000
B	3 weeks	$65,000
C	3 weeks	~~$58,666~~ → 68666
D	2 weeks	$41,500
E	2 weeks	$80,000

(1)原本成本:　54000 + 62000 + 66000 + 40000 + 75000 = 297000

間接成本:　5000 * 12 = 60000

原本總成本 =>　297000 + 60000 = **357000**

(2)現在趕工成本:　54000 + 65000 + 68666 + 41500 + 80000 = 309166

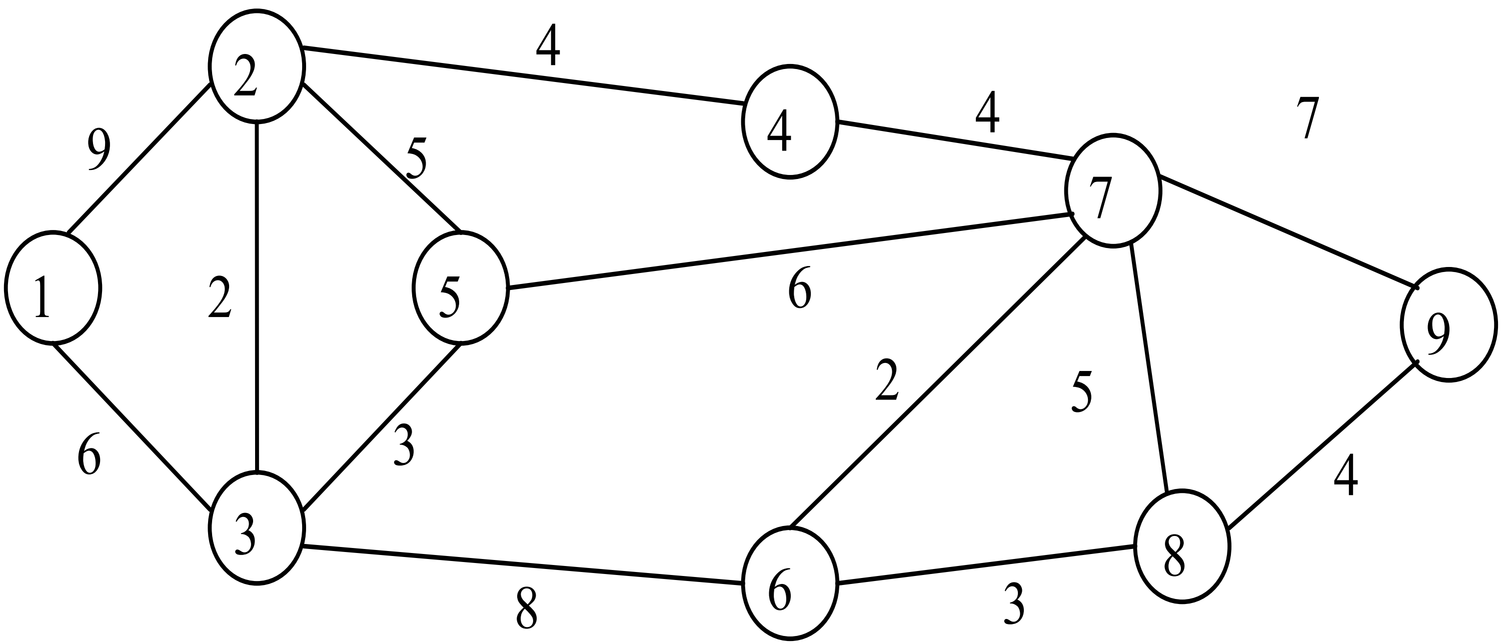
間接成本:　5000 * 8 = 40000

趕工的總成本 =>　299166 + 40000 = **349166**

因此 => 原本總成本 - 趕工的總成本 = 357000 – 349166 = 7834

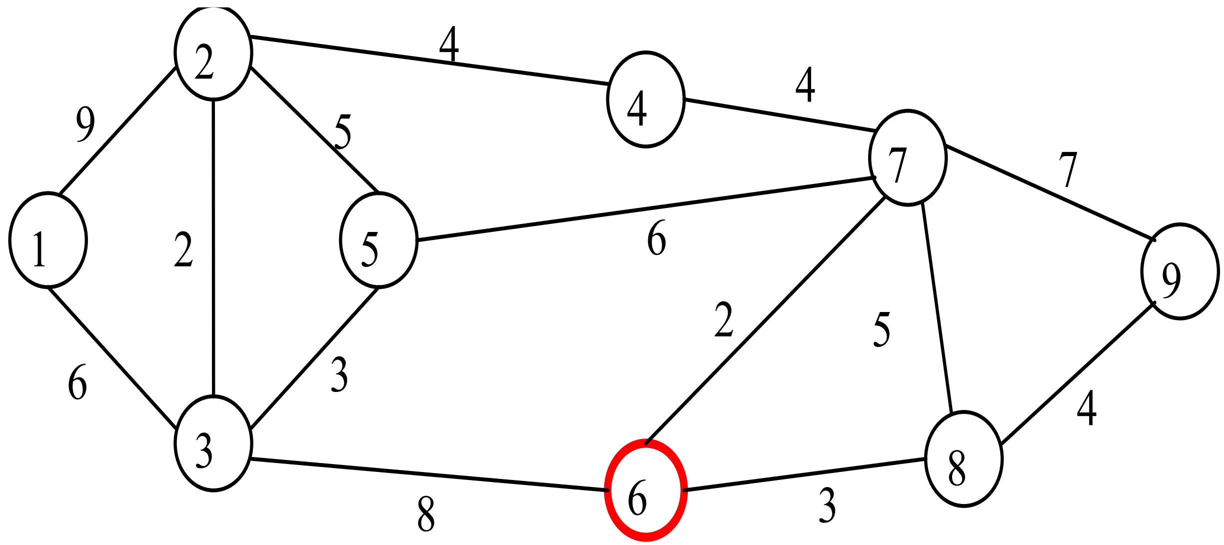
例題8-1

設有網路圖如下，代表各城市間的距離，為架設電腦網路，請求出最短伸展樹

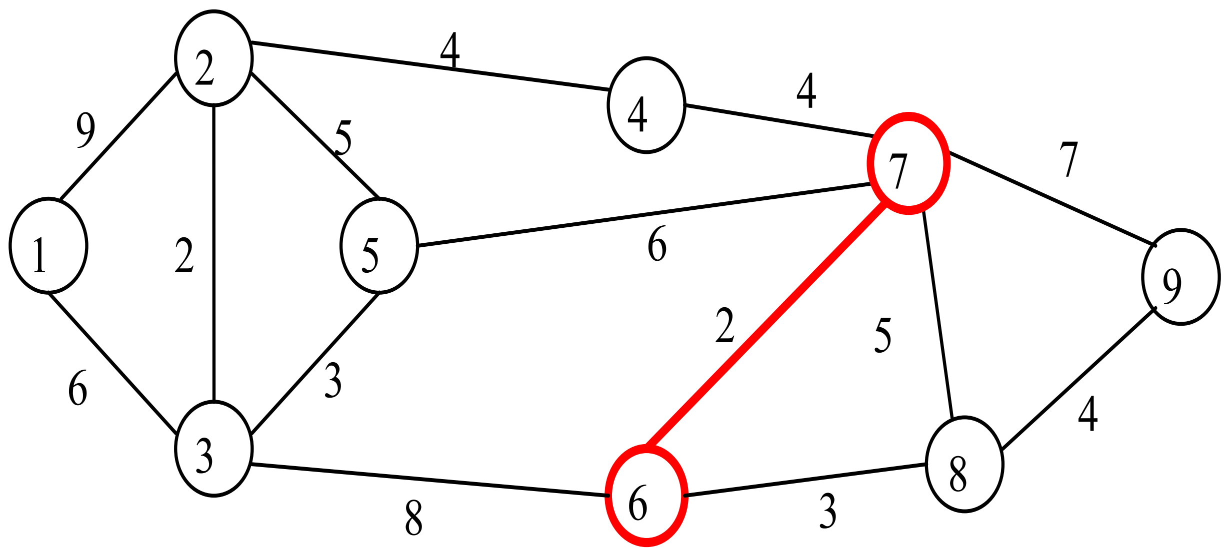


解：

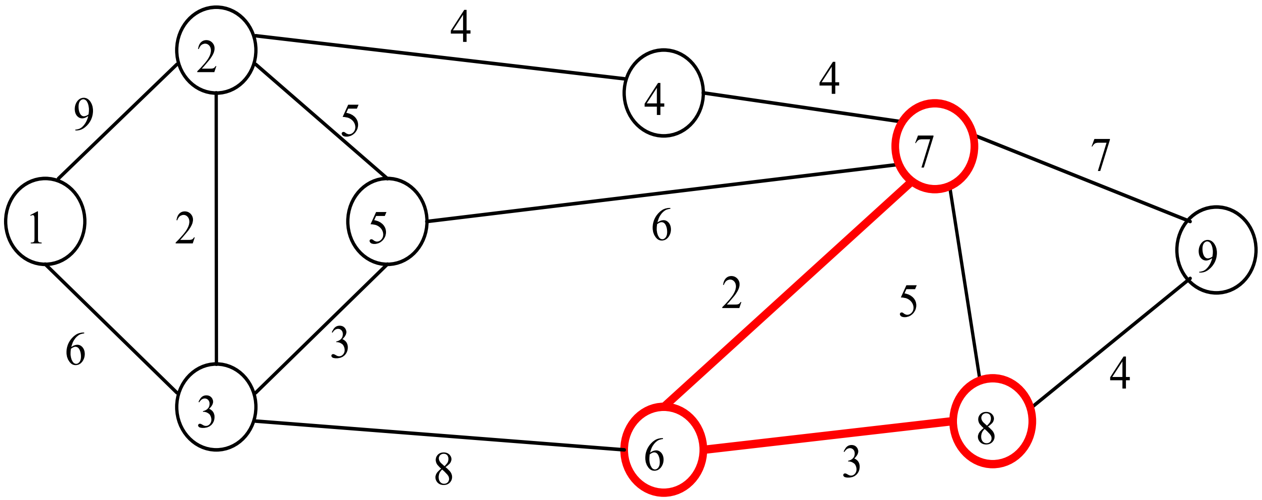
先隨機圈選節點⑥。與節點⑥相連的有節點③⑦⑧，其間弧線以節點⑥⑦間的最短，故再圈選⑥⑦間之弧線與節點⑦。再檢視與節點⑥⑦相連的有節點③④⑤⑧⑨，其間弧線以節點⑥⑧間弧線最短，因而圈選⑥⑧間弧線與節點⑧，如此反覆，即可得最佳解，請參見圖 8-4 (a) ~ (I)。



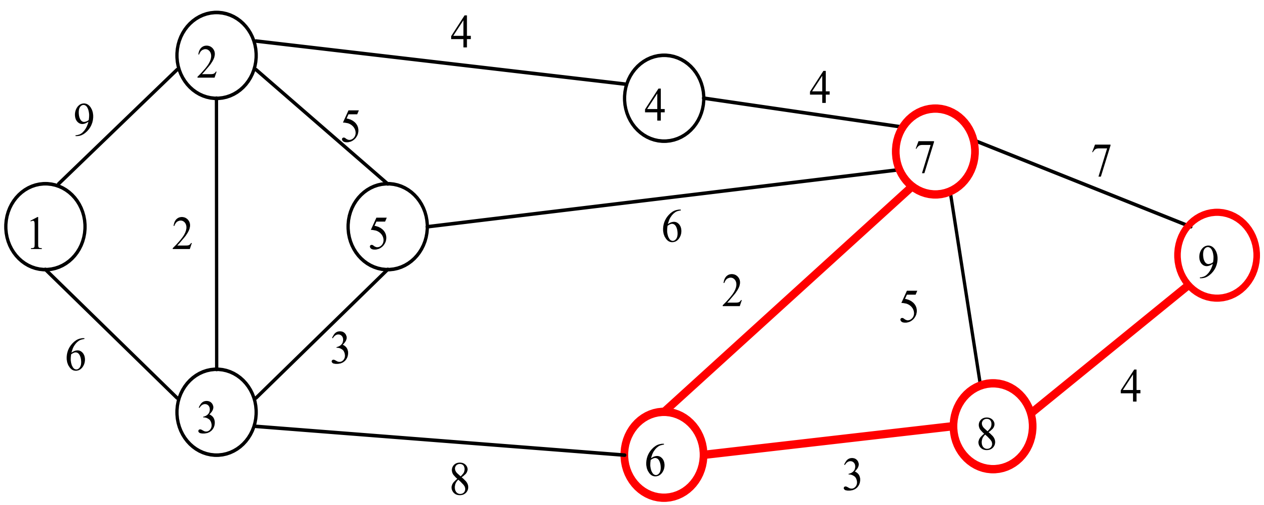
(a)



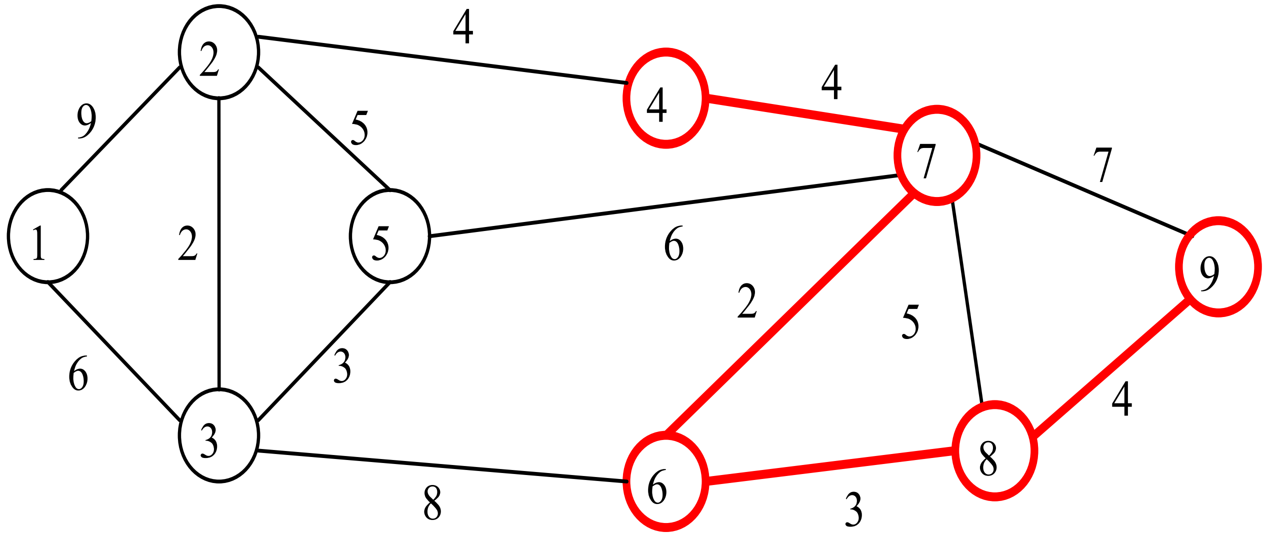
(b)



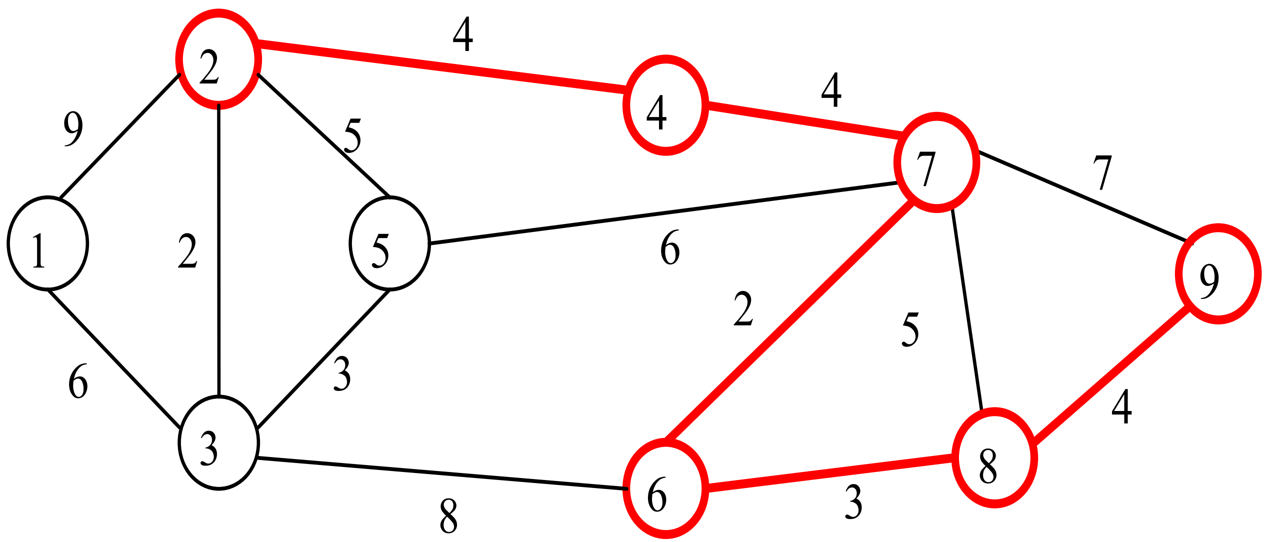
(c)



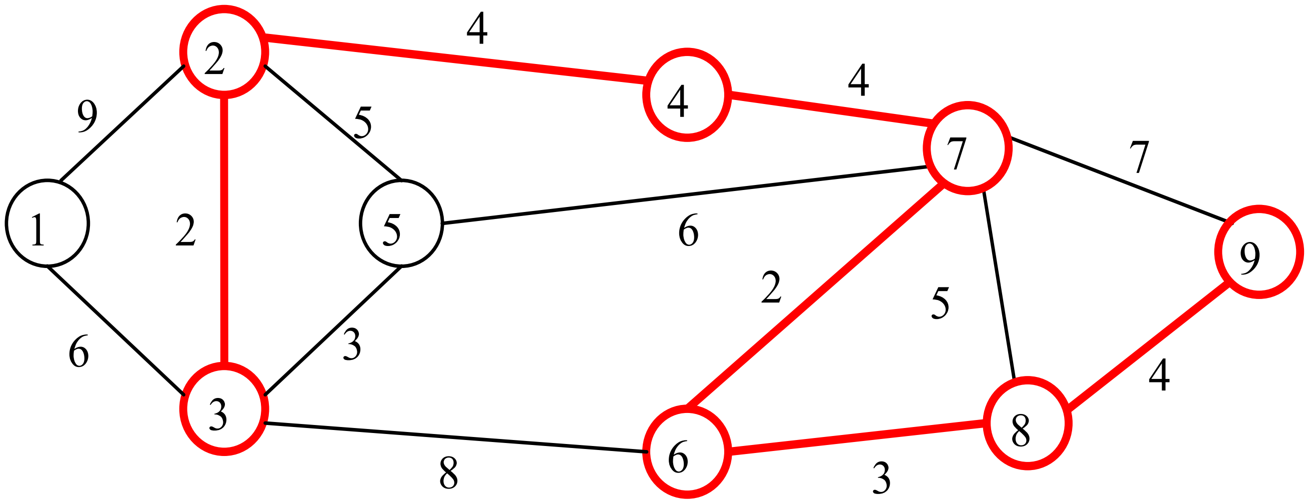
(d)



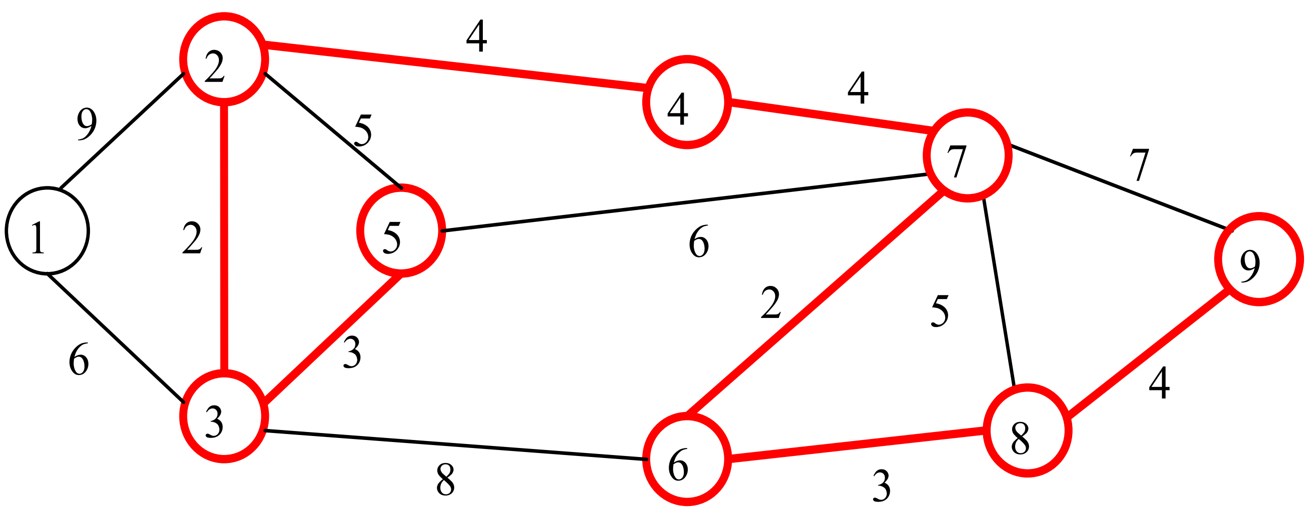
(e)



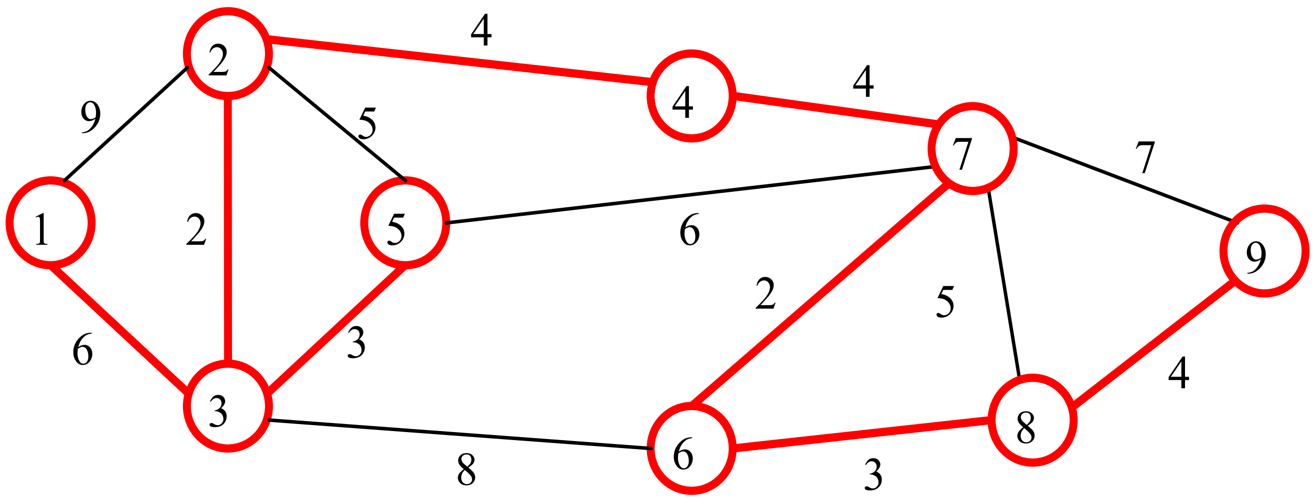
(f)



(g)



(h)

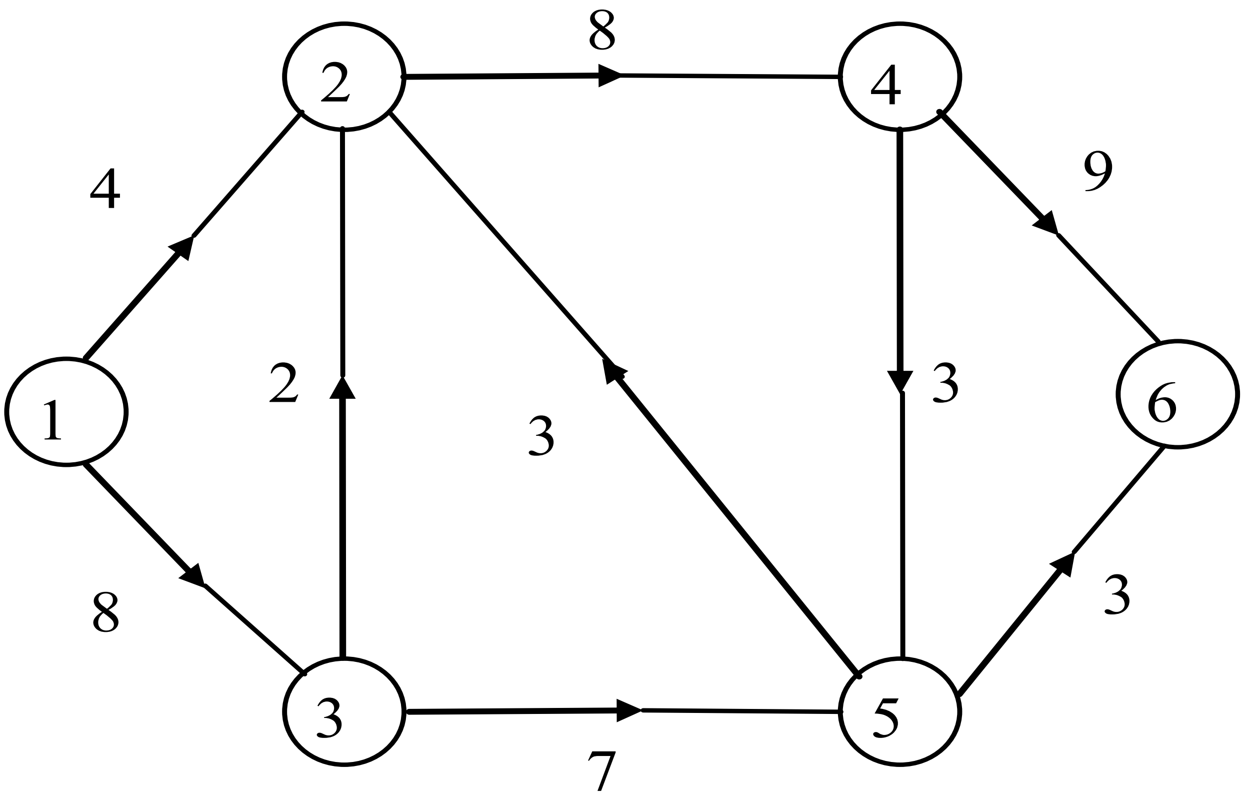


(I)

$$伸展樹之總長 = 6+2+3+4+4+2+3+4=28$$

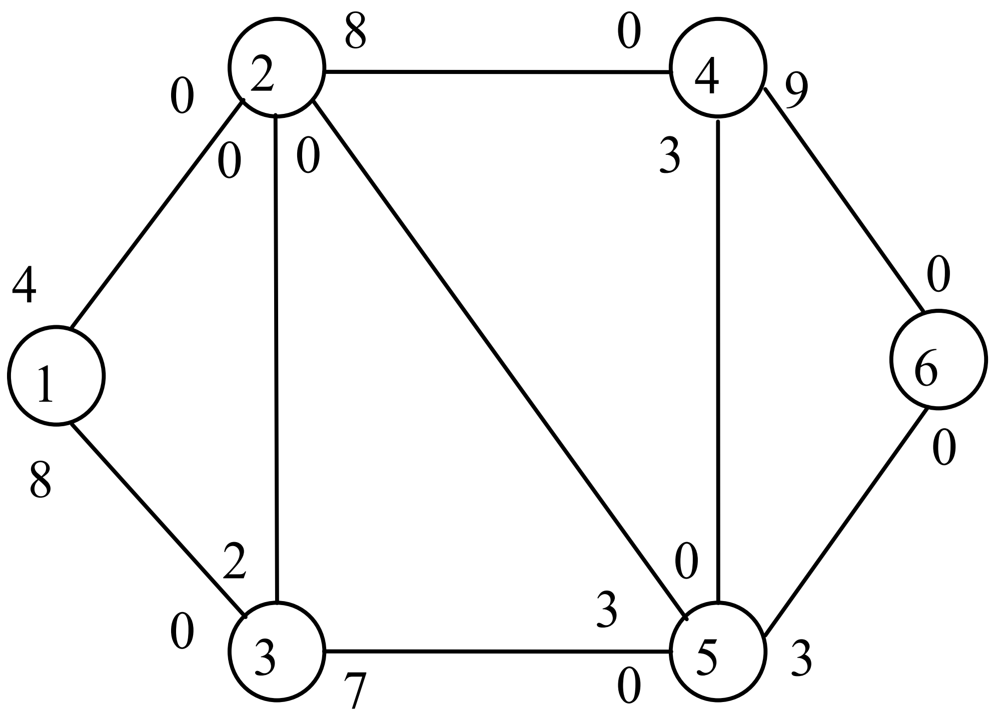
例題8-6

求網路自節點 ① 至節點 ⑥ 的最大流量



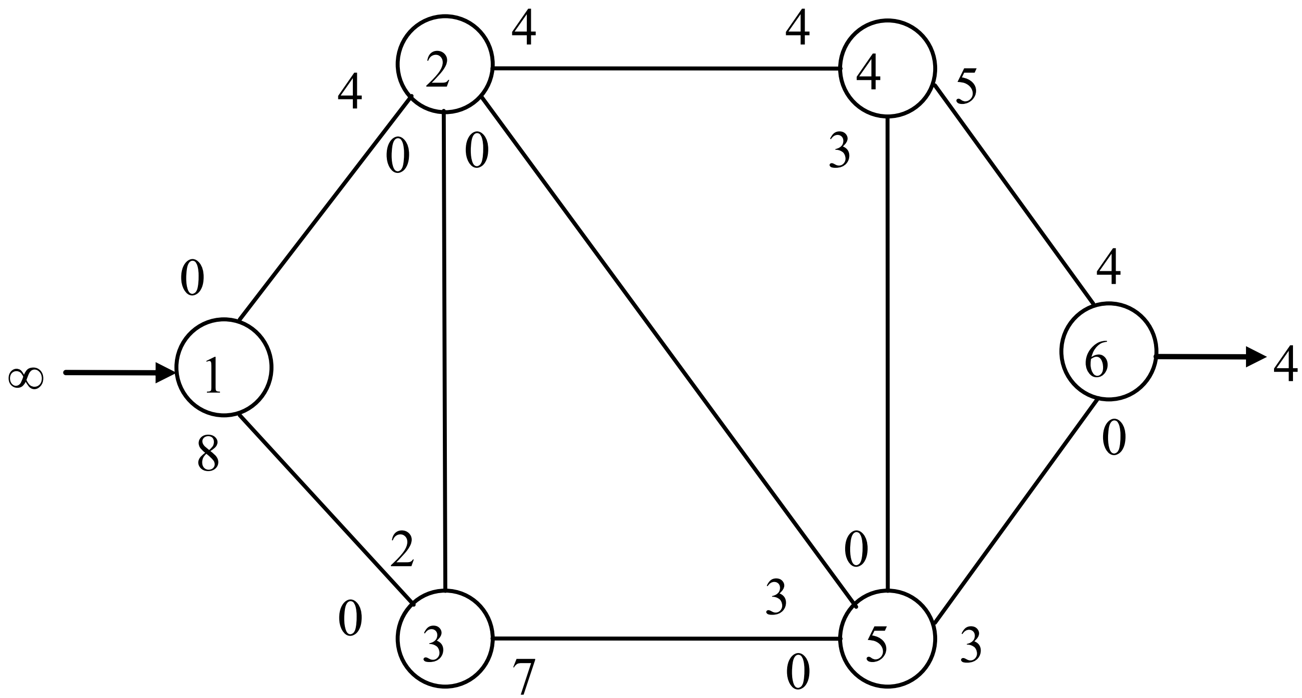
解：

依題目網路圖得下之網路圖。解題過程詳見圖 8-14(a) ~ (e)



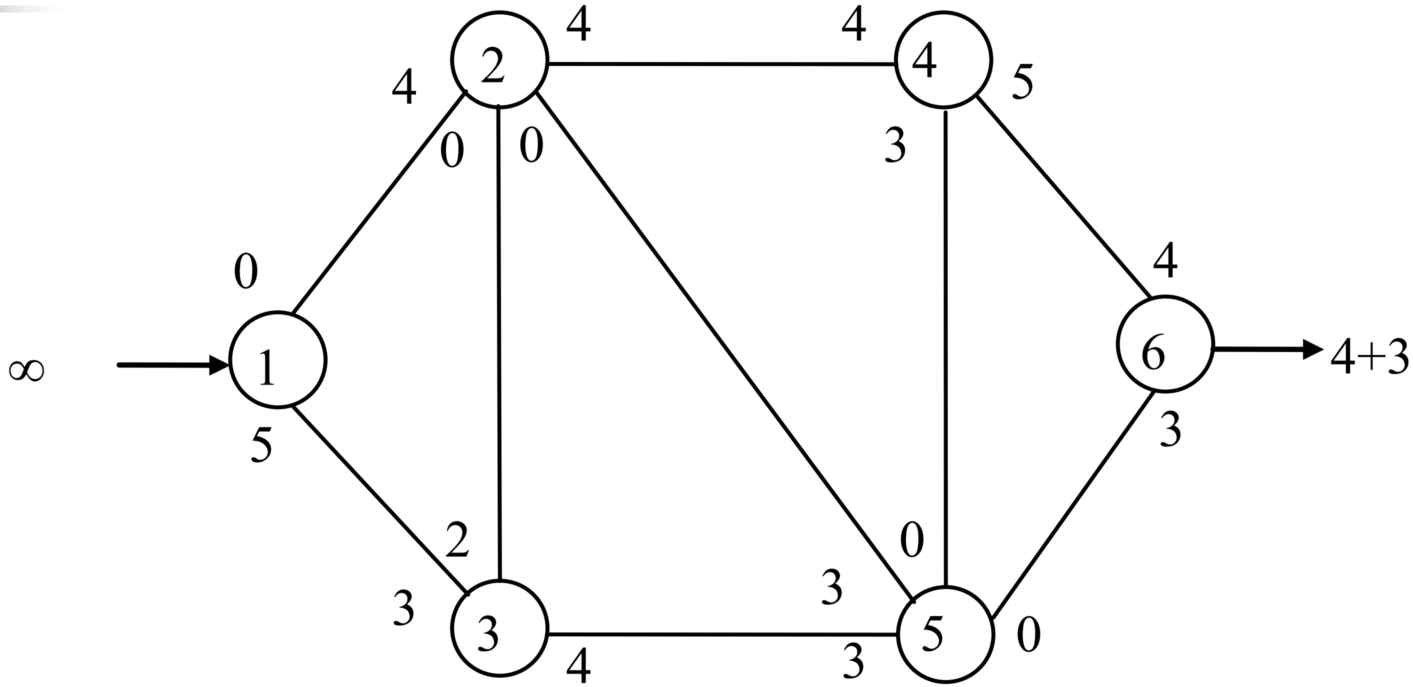
路徑 ① — ② — ④ — ⑥，$f^*=4$

(a)



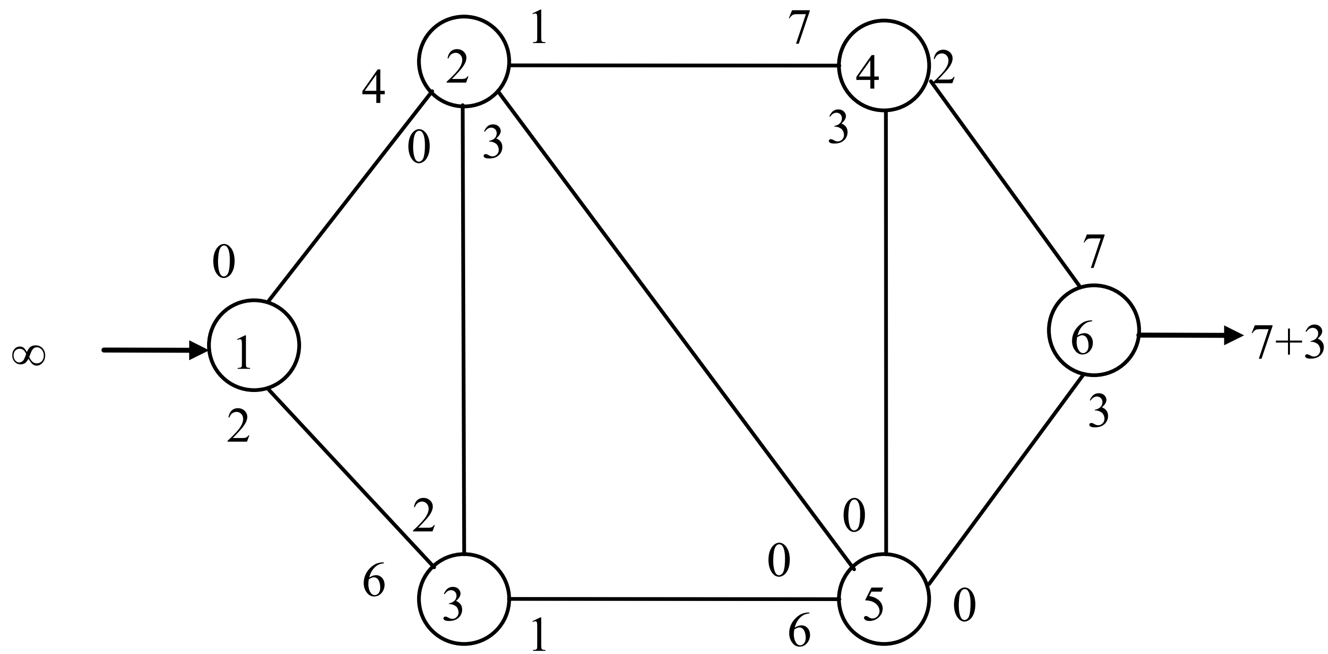
路徑 ①—③—⑤—⑥，$f^*=3$

(b)



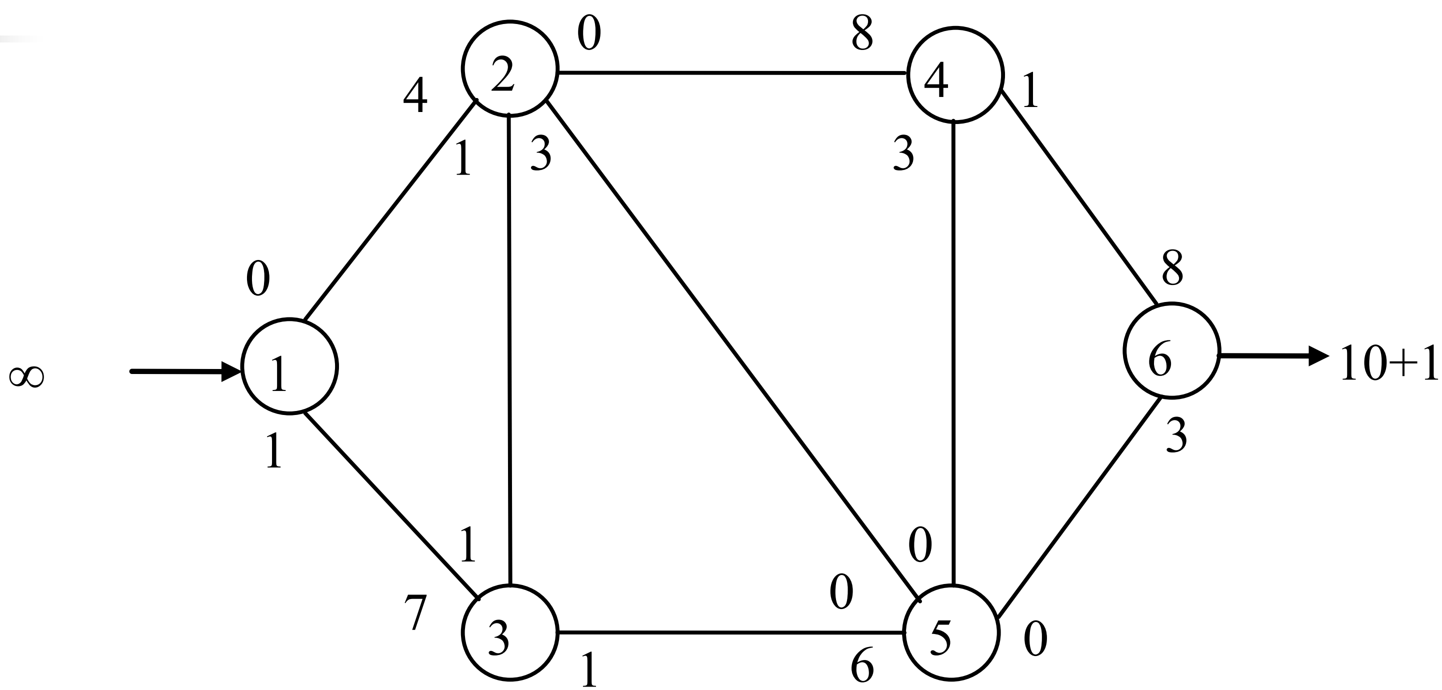
路徑 ①—③—⑤—②—④—⑥，$f^*=3$

(c)



路徑 ① — ③ — ② — ④ — ⑥ ，$f^*=1$

(d)



已無自節點 ① 到節點 ⑥ 的正流通量路徑，故最大流量 11。

(e)